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Course: INSY 8311 & Database Development with PL/SQL

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Problem Statement

Restaurants frequently face revenue leakage, inventory discrepancies, and slow customer service due to manual order-taking, fragmented paper receipts, and delayed sales reconciliation. Without an integrated system, managers struggle to track daily revenue in real time, monitor raw ingredient depletion accurately, or prevent billing discrepancies.

Objectives

Main Objective: Build a normalized Oracle PL/SQL database to automate restaurant order processing, enforce real-time inventory deduction, and provide live sales analytics.

Specific Objectives:

1. Design a 3rd Normal Form (3NF) relational database schema for branches, menu items, raw ingredients, recipes, staff, and sales orders.
2. Write PL/SQL triggers to automatically decrement raw ingredient stock levels upon order completion and block sales when inventory is insufficient.
3. Create PL/SQL procedures and packages for order entry, bill calculation, staff shift management, and sales audits.
4. Develop an interactive dashboard to monitor daily revenue KPIs, top-selling menu items, and low-stock alerts in real time.

Data Source

Dataset: Restaurant Customer Orders, Menu Pricing, and Sales Transactions Dataset (Kaggle)

Link: https://www.kaggle.com/datasets

Size: ~10,000+ records

License: Educational / Open Access

Planned Tables

1. `BRANCHES`: Stores branch locations, contact details, and operating status.
2. `MENU_ITEMS`: Central catalog detailing dish names, descriptions, categories, and unit prices.

3. `INGREDIENTS`: Tracks raw food items, measurement units, and current stock quantities.
4. `RECIPE_ITEMS`: Maps menu items to their required raw ingredients for automated stock depletion.
5. `STAFF`: Stores employee records, roles (Waiter, Chef, Cashier), and shift assignments.
6. `ORDERS`: Transaction headers recording order timestamps, table numbers, staff IDs, and order statuses.
7. `ORDER_DETAILS`: Line items linking specific orders to menu items and quantities.
8. `PAYMENTS`: Tracks payment methods (Cash, Mobile Money, Card) and transaction totals.

Planned PL/SQL Features

- Function (`fn_calculate_daily_revenue`): Computes and returns total daily sales revenue for a given branch and date.
- Procedure (`sp_process_new_order`): Handles order entry, item validation, subtotal calculations, and order status updates automatically.
- Trigger (`trg_deduct_inventory`): A compound trigger that automatically updates ingredient stock levels when an order is finalized and raises an application error if stock is insufficient.
- Package (`pkg_restaurant_ops`): Consolidates all order management, inventory validation, and sales reporting procedures into one encapsulated unit.

Innovation

Option: Analytics Dashboard & User Interface

Tool: Oracle APEX / Power BI

Description: A web-based operational interface and live visual dashboard connected to the Oracle database displaying daily revenue trends, top-selling menu items, and automated low-stock alerts.

References

Islam, M. A., et al. (2024). Restaurant Order Management System. ResearchGate.

National Institute of Statistics of Rwanda [NISR]. (2025). Rwanda Business Establishment Census. NISR Rwanda.

Patel, M., et al. (2025). Enhancing Restaurant Operations through a Modern POS System. International Journal of Innovative Research in Technology (IJIRT).

Oracle Corporation. (2026). Oracle Database PL/SQL Language Reference. Oracle Docs.